Oxyile Protocol - Detailed FinTech Enterprise Architecture

Actors
Platform Admin (Pitch Reviewer)
Investor
Borrower
reviews pitch/compliance + verifies collateral
JIT funding + role-based actions
reviews marketplace + funds escrow
HTTPS/JSON form submit
handshake/collateral actions
verification decisions + PDF export
onboards + submits waitlist answers
uploads collateral + links bank
Frontend Layer - Next.js on Vercel
Payment Success Page
/handshake/success
/payments/mandate-comp
Admin Portal
time collateral verifica
compliance preview,
PDF export)
Investor Marketplace
ple-aware handshake
IT Fund Escrow flow
Waitlist UI
Open Banking consen
o-applicant willingnes
lockchain importance
Borrower Dashboard
(Collateral upload,
loan requests,
mandate linking)
RLS-aware access checks
session validation
FLOW 1: verify/reject collateral
FLOW 2: click "Fund Escrow"
signed URL resolution
FLOW 4: waitlist compliance data capture
FLOW 1: collateral docs upload
upload docs path references
borrower-side status updates
API Entry Layer
Vercel Edge / Next AP
post-payment callbacks
export waitlist/pitch compliance profile
FLOW 4: pitch/compliance PDF export
Backend Core - Next.js API Routes & Server Actions
Escrow & JIT Funding Controller
EscrowAndRoute(han
Funding(handshakeI
ndshake State Machi
NG → FUNDED → A
FLOW 2: payout routing to borrower phase
BorrowerBankLink(ha
Admin Collateral Verification Services
ojectCollateralAsset(
dingCollateralVerifica
st/Profile PDF Gene
VerifyCollateralAsset(
oved_value, max_ltv
Identity & Compliance Router
C/AML + Role Validat
ase Auth Session Re
POST /api/waitlist
final on-chain anchor call
trigger relayer path
create/activate mandate & subscription
FLOW 2: funds held in escrow
checkout completion callback
FLOW 3: tx hash -> Supabase
create billing request + checkout URL
FLOW 2: GoCardless JIT checkout
persist PENDING/FUNDED/ACTIVE transitions
set collateral_status=rejected
read pending collateral queue
email confirmations + alerts
update verified values + max_ltv_amount
FLOW 4: store + expose for admin preview
insert waitlist + compliance preferences
policy-enforced SQL access
live pending verification updates
FLOW 1: realtime alert -> admin queue
handshake status live updates
Platform Reliability & Async (Optional Enterprise Layer)
ession/Response Cac
Monitoring/Alerting
Event Stream
rendered PDF artifact
Web3 Polygon Relayer
Relayer Service
.env.POLYGON_PRIVA
lated server-side us
Agreement Hashing
andshake + EMI ter
Transaction Submitte
(Amoy RPC)
confirmed tx hash
FLOW 3: hash agreement + mint tx
FLOW 3: success webhook/callback
broadcast tx
persist tx hash in handshakes
External APIs & Services
Polygon Amoy Testne
(Immutable ledger)
GoCardless API
(Open Banking,
JIT checkout,
client money escrow,
DD mandates)
Resend API
(Alerts, verifications)
Supabase - PostgreSQL + Realtime + Auth
Storage Buckets
(collateral docs)
Auth + RLS Policies
Core PostgreSQL
(users, profiles,
waitlist, handshakes,
collateral tracking,
andates, tx hashes)
postgres_changes feed
Realtime WebSockets
(postgres_changes)